How the data moves

Fictional source systems. Real data engineering concepts. One reconciled finance reporting pipeline.

01 GENERATE SOURCE EXPORTS LOCALLY WITH PYTHON

NetSuite-like

accounts / vendors / gl_lines
12 accounts / 16,648 GL lines

Salesforce-like

customers / opportunities
contracts / bookings
customer_monthly

Workday-like

employees / employee_monthly
96 employee records

Pigment-like

versions / financial_plan
headcount_plan

Shared master data

departments
6 consistent department IDs

02 RUN THE MANUAL REFRESH

Python ingestion

Freeze + validate input CSVs
Batch loads: WRITE_TRUNCATE
Verify every uploaded field

BigQuery raw

raw_netsuite / raw_salesforce
raw_workday / raw_pigment
raw_shared: 14 tables total

dbt Core + SQL

8 staging + 6 finance views
BigQuery executes the SQL
104 data tests + 4 unit tests

Verified export

4 reporting SQL queries
4,047 rows checked in Python
Saved warehouse snapshot

Data + Sites dashboard

P&L, budget, ARR, bookings
Workforce + department spend
Local app; separate publication

Independent reconciliation

Frozen CSVs -> local SQLite calculations
Python compares every reporting field
Dashed line = validation, not another source

Metric rules

Revenue / bookings / costs: sum the period
ARR / customers / headcount / FTE: ending value

GitHub stores code; Actions tests it

Python, SQL, dbt and dashboard source in Git
Manual CI: fresh synthetic data + offline tests
No Google credentials or warehouse refresh in CI

Separate API learning exercise

Mock HTTP API -> paginated Python extractor
130 customers / 6 pages -> local customers.csv
API-to-BigQuery connection remains deferred

WHAT CHANGES WHEN YOU REFRESH?

Replaces raw data, rebuilds views, checks results, updates the local snapshot.
Hosted publication is separate. Browsing the dashboard does not query BigQuery.